Dr. Kumar Sonu

Curriculum vitae

Name: **Dr. Kumar Sonu**

1. **Gender:** Male
2. **E-mail:** kumar_sonuasg@yahoo.com
3. **Vidwan-ID: 229569**
4. **Mobile: 6378665397**
5. **Google Scholar id:** https://scholar.google.com/citations?user=
6. **Date of Birth & Sex:** 02/04/1991 & Male
7. **Permanent Address**

S/o Mr. Ganesh Kumar

House No. 9, Amokhari,

P.O. Bargaon, Dist. Nawada

Bihar - 805122, India

8. **Qualifications:** BE-Mechanical Engineering, M Tech-Energy and Environmental Engineering, PhD.

S.No.	Degree	Institution	Year	Division/Class
1.	PhD (Civil Engineering)	Manipal University Jaipur	2021	----
2.	M. Tech Energy & Environmental Engineering	VIT, Vellore	2016	First class Distinction
3.	BE-Mechanical Engineering	Visvesvaraya Technological University, Belgaum	2013	First class Distinction

9. **Employment Experience**

S.No.	Position & Organization	Nature of Job	Period
1	Senior Project Engineer, Indian Institute of Technology , Kanpur	Project administration	------
2	Associate Professor, HoD-ME/ Asst. Dean (R&D) Kashi Institute of Technology, Varanasi	Teaching, administration and research	4 years
3	Assistant Manager, Shree ram transport finance company Ltd	Branch Management	1 year

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10. List of Publications

Journal Publications (37)

1. **Sonu, K**., Syed, Z., Singh, G. et al. (2025). Performance evaluation of a Dracaena plant-based microbial fuel cell utilizing municipal solid waste compost as substrate and stainless steel mesh-supported 3D Cassia Fistula as bioanode. Waste Dispos. Sustain. https://doi.org/10.1007/s42768-025-00244-3
2. **Sonu, K**., Sogani, M., & Syed, Z. (2025). Plant microbial fuel cell performance assessment utilizing anode from Delonix regia fruit pod. *Biomass Conversion and Biorefinery*, 1-8.
3. Maheshwari, K., Sogani, M., Syed, Z., Sen, H., Rajvanshi, J., **Sonu, K**., ... & Agrawal, P. (2025). Microbial desalination cells in brackish RO reject treatment: Challenges and opportunities. Desalination, 119022.
4. Maheshwari, K., Sogani, M., **Sonu, K**., & Syed, Z. (2025). Enhanced Microbial Desalination Cell Performance Utilizing Polyurethane-Modified Bio-Based Polymeric Membrane. *ChemistrySelect*, *10*(8), e202500044.
5. **Sonu, K**., Sogani, M., Syed, Z., Maheshwari, K., Rajvanshi, J., & Sengupta, N. (2025). Effectiveness of a Biofertilizer in the Treatment of Carpet Industry Dye Wastewater in an Earthen Pot–Coupled Plant Microbial Fuel Cell. *Journal of Hazardous, Toxic, and Radioactive Waste*, *29*(2), 04024050.
6. **Sonu, K**., Sogani, M., Syed, Z., Maheshwari, K., Rajvanshi, J., & Sengupta, N. (2024). Rice husk-modified ceramic membrane for microbial desalination cell. *Biomass Conversion and Biorefinery*, 1-8.
7. **Sonu, K**., Sogani, M., Syed, Z., Maheshwari, K., Rajvanshi, J., & Sengupta, N. (2024). Performance Investigation of Aloe Vera Plant-Based Microbial Fuel Cell Using Anode Constructed of Carbonized Ipomoea Carnea. *Waste and Biomass Valorization*, 1-11.
8. Rajvanshi, J., Sogani, M., Tziouvaras, G., Kumar, A., Syed, Z., **Sonu, K**., ... & Sen, H. (2024). An analytical review on revamping plastic waste management: exploring recycling, biodegradation, and the growing role of biobased plastics. *Environmental Science and Pollution Research*, 1-19.
9. **Sonu, K**., Sogani, M., Syed, Z., Maheshwari, K., Rajvanshi, J., & Sengupta, N. (2024). Performance investigation of Aloe vera plant-based microbial fuel cell using anode constructed of carbonized Ipomoea carnea.
10. **Sonu, K**., Sogani, M., Syed, Z., Rajvanshi, J., & Pandey, S. C. (2024). Performance evaluation of Epipremnum aureum plant-based microbial fuel cell using composite anode made up of carbonized corncob and carbon rod. *Biomass Conversion and Biorefinery*, *14*(4), 5149-5156.
11. Mehta, S., Sogani, M., Syed, Z., **Sonu, K**., Kumar, A., & Vyas, A. D. (2024). River rejuvenation in urban India for enhancing living conditions through integrated water resources management. *Environmental Science and Pollution Research*, *31*(1), 682-698.
12. **Sonu, K**., Sogani, M., Syed, Z., & Rajvanshi, J. (2024). Improved degradation of dye wastewater and enhanced power output in microbial fuel cells with chemically treated corncob anodes. *Biomass Conversion and Biorefinery*, *14*(1), 375-386.
13. Syed, Z., Sogani, M., Rajvanshi, J., & Sonu, K. (2023). Microbial biofilms for environmental bioremediation of heavy metals: a review. *Applied Biochemistry and Biotechnology*, *195*(9), 5693-5711.
14. Rajvanshi, J., Sogani, M., Kumar, A., Arora, S., Syed, Z., **Sonu, K**., & Kalra, A. (2023). Perceiving biobased plastics as an alternative and innovative solution to combat plastic pollution for a circular

economy. *Science of The Total Environment*, *874*, 162441.

15. **Sonu, K**., Sogani, M., Syed, Z., Rajvanshi, J., Sengupta, N., & Kumar, P. (2023). The effects of waste jasmine flower as a substrate in a single chamber microbial fuel cell. *Biomass Conversion and Biorefinery*, 1-9.
16. Sogani, M., **Sonu, K**., Syed, Z., & Rajvanshi, J. (2023, March). Preparation of biofertilizer blend from banana peels along with its application in agriculture and plant microbial fuel cell. In *IOP Conference Series: Earth and Environmental Science* (Vol. 1151, No. 1, p. 012034). IOP Publishing.
17. **Sonu, K**., Sogani, M., Syed, Z., Rajvanshi, J., & Sengupta, N. (2022). Effectiveness of rice husk in the removal of methyl orange dye in Constructed Wetland-Microbial Fuel Cell. *Bioresource Technology Reports*, *20*, 101223.
18. **Sonu, K**., Sogani, M., Syed, Z., & Rajvanshi, J. (2022). Performance evaluation of Tagetes erecta plant microbial fuel cell using the composite ceramic anode of rice-husk, mild-steel dust, and soil. *Fuel Cells*, *22*(5), 197-204.
19. Syed, Z., Sogani, M., Sharma, G., **Sonu, K**., Rajvanshi, J., & Gupta, N. S. (2022). Framework to improve biohydrogen generation with estrogen co-metabolism under complete suppression of nitrogen source. *Bioresource Technology*, *360*, 127595.
20. **Sonu, K**., Sogani, M., Syed, Z., & Rajvanshi, J. (2022). The effects of wheat and rice straw as a substrate on the treatment of reverse osmosis reject wastewater in a single chamber microbial fuel cell. *ChemistrySelect*, *7*(18), e202103924.
21. Syed, Z., Sogani, M., Kumar, A., Rajvanshi, J., Sharma, G., & **Sonu, K**. (2022). Biodegradation of synthetic estrogen using bioelectrochemical system and degradation pathway analysis through Quadrupole-time-of-flight-mass spectrometry. *Bioresource Technology*, *349*, 126857.
22. **Sonu K**, Sogani M, Syed Z, Rajvanshi J (2021). Improved degradation of dye wastewater and enhanced power output in microbial fuel cells with chemically treated corncob anodes. Biomass Conversion and Bio refinery.
23. Syed, Z, **Sonu, K**., Sogani, M. (2021). Cattle Manure Management using Microbial Fuel Cells for Green Energy Generation. *Biofuels, Bio-products and Bio-refining*, https://doi.org/10.1002/bbb.2293
24. **Sonu, K**., Sogani, M., & Syed, Z. (2021). Integrated Constructed Wetland-Microbial Fuel Cell using Biocharas Wetland Matrix: Influence on Power Generation and Textile Wastewater Treatment. *ChemistrySelect*, *6*(32), 8323-8328.
25. **Sonu, K**., Sogani, M., & Sharma, M. K. (2021). Performance evaluation of plant microbial fuel cell with Epipremnum aureum plant using composite anode made of corn cob and carbon rod. *International Journal of Environmental Studies*, 1-12.
26. Syed Z, Sogani M, Dongre A, Kumar A, **Sonu K**, Sharma G, Gupta A B. Bioelectrochemical systems for environmental remediation of estrogens: A review and way forward. *Science of The Total Environment*. 2021 Mar 18:146544.
27. Syed, Z., **Sonu, K**., Dongre, A., & Sogani, M. (2021, June). Assessing the Surface Water Quality of Ana Sagar Lake and its Bioremediation in Modified Constructed Wetland. In *IOP Conference Series: Earth and Environmental Science* (Vol. 796, No. 1, p. 012026).

28. **Sonu K**, Sogani M, Syed Z, Dongre A, Sharma G. Improved decolorization of dye wastewater and enhanced power output in the electrically stacked microbial fuel cells with H_2O_2 modified corncob anodes. *Environ Prog Sustainable Energy*. 2021; ep13638. https://doi.org/10.1002/ ep.13638.
29. **Sonu, K**., Sogani, M., Syed, Z., Dongre, A., & Sharma, G. (2020). Enhanced Decolorization and Treatment of Textile Dye Wastewater Through Adsorption on Acid Modified Corncob Derived Biochar. *ChemistrySelect*, *5*(39), 12287-12297.
30. **Sonu, K**., Sogani, M., Syed, Z., Dongre, A., & Sharma, G. (2020). Effect of Corncob Derived Biochar on Microbial Electroremediation of Dye Wastewater and Bioenergy Generation. *ChemistrySelect*, *5*(31), 9793-9798.
31. Syed, Z., **Sonu, K**., Dongre, A., Sharma, G., & Sogani, M. (2020). A review on Hydroxamic Acids: Widespectrum Chemotherapeutic Agents: *International Journal of Biology and Biomedical Engineering.* Volume 14. DOI: 10.46300/91011.2020.14.12.
32. **Sonu, K**., Syed, Z., & Sogani, M. (2020). Up-scaling microbial fuel cell systems for the treatment of real textile dye wastewater and bioelectricity recovery. *International Journal of Environmental Studies*, 1-11. DOI: 10.1080/00207233.2020.1736438.
33. **Sonu K**, Syed Z, Sogani M, (2019) Microbial Fuel Cell in Domestic Wastewater Treatment Plant – An Innovative Step towards Energy Generation and Waste Reduction. *J Sci Ind Res* Vol 78, August 2019: 555-557.
34. Syed Z, Sogani M, **Sonu K**, Dongre A (2019) Bioelectrical Systems for Removal of Endocrine Disrupting Compounds from Waste Water: A Review. *Global Journal of Engineering Science and Researches*6(2): February 2019 doi: 10.5281/zenodo.2558191
35. Chaithanya, M. S., Thakur, S., Sonu, K., & Das, B. (2017, November). Preliminary investigation of single chamber single electrode microbial fuel cell using sewage sludge as a substrate. In *IOP Conference Series: Materials Science and Engineering* (Vol. 263, No. 3, p. 032008).
36. **Sonu, K**. (2017). Generation of bio-electricity from sewage sludge using single chamber microbial fuel cell. *Journal of Environmental Science and Public Health*, *1*(2), 68-74.
37. **Sonu, K**., & Das, B. (2016). Comparison of the output voltage characteristics pattern for sewage sludge, kitchen waste and cow dung in single chamber single electrode microbial fuel cell. *Indian Journal of Science and Technology*.

Conference Presentations (13)

1. **Sonu K**, National conference on Innovation in Biotechnology and Bioengineering on 27th April 2024 at Kashi Institute of Technology, Varanasi.
2. **Sonu K**, Sogani M, Syed Z , 9th International Conference on Advances in Energy Research (ICAER) at IIT Bombay, 08 Dec, 2023-10 Dec, 2023

3. **Sonu K** (2023) ICC Water & Wastewater Innovation Conference hosted by Indian Institute of Technology Kharagpur.- September 29-30, 2023.
4. Syed Z, **Sonu K**, Dongre A, Sogani M (2021) Assessing the Surface Water Quality of Ana Sagar Lake and its treatment via Bioremediation. "Community Based Research and Innovations in Civil Engineering (CBRICE) 2021", Manipal University Jaipur, India, March 18-19, 2021.
5. **Sonu K**, Sogani M, Syed Z (2020) Alkaline Modification of Used Carbon-Rod Anodes for Restoring Power Generation by Human Urine in Microbial Fuel Cells, 1st Virtual International Conference on Advancement in Nanotechnology, Kingston Engineering College, Vellore, Tamil Nadu, India, September 11-12, 2020.
6. Syed Z, **Sonu K**, Dongre A, Sogani M (2020) Isolation and Characterization of a Novel Electrogenic Microbe: Potential Candidate for use in Microbial Fuel Cell. "SPARC Indo-Belgium workshop", IIT Kharagpur, India, February 26-27, 2020.
7. **Sonu K**, Syed Z, Dongre A, Sogani M (2019) A Cost Effective Small Scale Microbial Fuel Cell System for Bio-electricity Generation with Simultaneous Dye Wastewater Treatment, "25th National and 3rd International ISHMT-ASTFE, Heat and Mass Transfer Conference", IIT Roorkee, Uttarakhand, India, December 28-31, 2019.
8. **Sonu K**, Syed Z, Dongre A, Sogani M (2019) Effect of Bio-char on Real Textile Dye Wastewater Treatment and Energy Production in a Microbial Fuel Cell, "International Exergy, Energy and Environmental Symposium", SRM, Chennai, Tamil Nadu, India, July14-18, 2019. ISBN: 978-93-5382-239-2.
9. Syed Z, Dongre A, **Sonu K**, Sogani M (2019) Microbial Fuel Cells – An Emerging Technology for Wastewater Treatment. "National Conference on Biotechnology and Environment for Sustainable Development" Dr. B. Lal Inst. of Biotechnology, Jaipur, India, March 29-30, 2019.
10. **Sonu K**, Syed Z, Dongre A, Sogani M, Sharma M K, (2019) Reuse of Solid Waste Material in Microbial Fuel Cell System for Bioelectricity Generation with Simultaneous Dye Wastewater Treatment. "National Conference on Biotechnology and Environment for Sustainable Development" Dr. B. Lal Inst. of Biotechnology, Jaipur, India, March 29-30, 2019.
11. **Syed Z**, Sogani M, Dongre A, Sonu K (2018) Bioelectrical Systems for removal of Bioactive Chemicals from Wastewater. *"International Conference of Waste Management 2018"*, Manipal University Jaipur, Jaipur, India, October 1-2, 2018.
12. Dongre A, **Sonu K**, Sogani M, Vyas A D (2016) Generation of Electricity from RO concentrate Using Microbial Fuel Cell. National Symposium on "*Water Purification and Reject Management (WPRM-2016)*", MNIT, Jaipur (India), November 12, 2016.
13. **Sonu K** (2016) Comparison of the output voltage characteristic pattern for sewage sludge, kitchen waste & cow dung in single chamber single electrode microbial fuel cell at controlled temperature, "2nd International Conference On Sustainable Energy and Built Environment", VIT University, Vellore, India, March 10-12, 2016.

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Patents filed/Granted with details (3)

1. **Sonu K**, Sogani M (2021) Microbial Fuel cell- syringe type (Indian Design No. 351229-001). The IndianPatent Office Kolkata. 13 Oct. 2021.
2. Sogani M, **Sonu K** (2021) Bioelectroremediation of Textile Dye Effluent: Using Cow Dung and Corncob Anode (India Patent Application No.202011002321). The Indian Patent Office Kolkata. 23 July 2021.
3. Sogani M, **Sonu K** (2020) Microbial Fuel Cell with Corn-cob Anode (Indian Design No. 323304-001). TheIndian Patent Office Kolkata. 09 Sept. 2020.

Books Published/Chapters contributed (7)

1. **Sonu, K**., Sogani, M., Syed, Z., & Rajvanshi, J. (2024). Treatment of Textile Dye Wastewater with Simultaneous Bio-electricity Generation in Microbial Fuel Cell: A Review. *Microbial Remediation of Hazardous Chemicals from Water & Industrial Wastewater Treatment Plant*, 219-243.

2. **Sonu, K**., Sogani, M., Maheshwari, K., Rajvanshi, J & Syed, Z. (2024) Performance Analysis of a Microbial Fuel Cell Using Solid Waste as Substrate in Terms of Various Parameters. *Advances in Clean Energy and Sustainability*, 411–416

3. Syed, Z., Sogani, M., Rajvanshi, J., & **Sonu, K** (2024). Use of industrial waste to synthesize novel electrode and membrane materials for microbial electrochemical technologies. doi: 10.2166/9781789063813_0223

4. Syed, Z., Sogani, M., Rajvanshi, J., & **Sonu, K**. (2022). Electroactive biofilm and electron transfer in microbial electrochemical systems. *Scaling Up of Microbial Electrochemical Systems*, 29-48.

5. Dongre, A., Sogani, M., **Sonu, K**., Syed, Z., Sharma, G. (2020). Treatment of Dairy Wastewaters: Evaluating Microbial Fuel Cell Tools and Mechanism. *Environmental Change and Sustainability* (Edited by Dr. Suriyanarayanan Sarvajayakesavalu and Dr. Pisit Charoensudjai) Intech Open. DOI: 10.5772/intechopen.93911
6. Sogani M, **Sonu K**, Dongre A (2017) Treatment of Industrial Wastewater Containing Amides Using Novel Bacterium in Semi-Continuous Reactor. *Civil Engineering Notes: Frontiers in Wastewater Treatment and Modelling.* (Edited by G. Mannina, University of Palarmo, Italy), Springer International Publishing, Pg. 354-359, ISBN: 978-3-319-58420-1

12. Research Project (1)

Providing low-cost solution and appropriate management framework for the reject disposal of Community-based RO Plants in many areas of Rajasthan. Role: Co principal, Year 2022, Amount ₹ 8000000/-, DST.

13. Course Attended (2)

1. Biomass Conversion and Biorefinery, NPTEL, 16 Jan, 2022-16 Apr, 2022
2. RS & GIS Applications in Water Resource Management, ISRO, 22nd May-09 through June-2017

14. Workshop Attended/Conducted (13)

1. Meta Materials and its Applications, 18 Jan, 2024-18 Jan, 2024
2. Importance of IPR-12 Jan, 2023- 12 Jan, 2023
3. ACADAMIC WRITING, 12 Apr, 2023-12 Apr, 2023
4. Next Generation Sequencing – From Genomics to Medicine, 25 Jul, 2022-07 Aug, 2022

5. Entrepreneurship, Research Methodology and Patent Writing, 25 Aug, 2022-29 Aug, 2022
6. X-ray Diffraction: Principle and Applications, 6th July, 2021 under Academia-Industry Symbiotic Alliance (AISA)
7. Innovation commercialization for Aatmrit Bharat, 14 Aug, 2020
8. Waste to bioenergy , June 28 to 4th July 2020, Sharda University & Maharashtra Institute of Technology
9. Universal Human Values and Professional Ethics,01 Jan, 2020- 07 Jan, 2020
10. Universal Human Value, BTU, Bikaneer from 5th July, 2019 through 12th July.
11. Sutras of Effective teaching and workplace Ethics, 25 Dec, 2018-26 Dec, 2018
12. Document Typesetting and Processing using LATEX, Department of Civil Engineering, Manipal University Jaipur, 25th November, 2017.
13. Universal Human Value, ANAND-ICE, Jaipur from 26th December,2017 through 2nd January, 2018

15. Peer Review (3)

1. Chemosphere,
2. National Academy Science letters
3. , International Journal of Environmental Analytical Chemistry

16. Membership IEEE -101526480

17. Experience of Administrative Responsibilities in Academic Institutions

Sr. No.	From	To	Administrative Position	Major Responsibility
1	22-08-2022	10-6-2024	IIC-President	R&D/Innovation/ StartUP
2	17-08-2023	10-6-2024	PMKVY-Coordinator	Organizing activates
3	17-08-2023	10-6-2024	Dean R&D	Organizing R&D activates
4	12-12-2020	17-08-2023	Head of Department	Department management
5	08-08-2021	13-06-2023	NAAC accreditation Criteria Head	NAAC documentation

18. Professional References

Name: Dr. Monika Sogani Designation: Associate Professor Organization: Manipal University Jaipur Email:monika.sogani@jaipur.manipal.edu Mobile: 9414207450	Name: Dr. Manoj Kumar Tiwari Designation: Associate Professor Organization: Indian Institute of Technology , Kanpur Email: mktiwari@iitk.ac.in Mobile: 9002491978

Dr. Kumar Sonu

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19. Declaration:

I declare that the foregoing information is correct and complete to the best of my knowledge and belief and nothing has been concealed/distorted.

Date: Name: Dr. Kumar Sonu

Place: Signature : ……

Dr. Kumar Sonu

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